

Markus Gruber

Backend Engineer | AI Systems

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Senior Backend Engineer with 9+ years of experience building scalable backend systems, distributed services, and AI-powered production systems using Python and C++. Experienced in secure software design, service integration, automation, and reliable data-processing platforms. Strong background in AI system integration, performance optimization, and developing software for demanding production environments.

I combine strong backend engineering skills with experience in AI systems, automation, performance optimization, and secure software design. My focus is on building maintainable platforms with reliable data processing, robust service integration, and predictable system behavior at scale.

Core Expertise

- Backend Development & System Architecture
- Distributed Systems & Service Integration
- Python Backend Engineering & Automation
- AI Systems & Machine Learning Integration
- Secure Software Development & System Reliability
- Performance Engineering & Scalability

Berufserfahrung

Mindbreeze GmbH, Software Engineer C++/Python

09/2022 – present

- Designed and developed scalable backend components, services, and automated workflows for an enterprise search platform processing complex information workloads.
- Developed Python-based tools and services for processing large datasets, automation, validation, and quality assurance.
- Developed reliable enterprise systems with focus on data integrity, secure processing workflows, controlled service interaction, and predictable production behavior.
- Improved production systems regarding performance, scalability, resource consumption, and reliability.
- Contributed to software architecture decisions and continuously improved development and testing processes.

Untis GmbH, Software Engineer C++

02/2022 – 06/2022

- Maintained and enhanced a complex MFC desktop application within a large legacy codebase.
- Investigated and resolved stability, performance, and maintainability issues.

Logicals, Software Engineer C

02/2021 – 01/2022

- Developed embedded software for safety-critical industrial applications under functional safety requirements.
- Implemented software architectures and testing strategies to ensure reliability of system-critical components.

EYYES, Software Engineer C++/Python

04/2017 – 12/2020

- Developed and deployed deep-learning-based computer vision systems for safety-critical real-time applications, including integration of AI models into production software environments with strict reliability requirements.
- Built automated data and machine-learning pipelines including data preparation, model training, evaluation, and integration of production AI models.

- Optimized training and inference pipelines regarding performance, scalability, and robustness.

Technical Skills

Programming Languages

Python, Go, C++, SQL, C, Rust, Bash

Backend & Distributed Systems

Backend Development, Distributed Systems, Microservices, REST APIs, Service Integration, Software Architecture, Performance Engineering

Security & Reliability

Secure Software Development, Authentication & Authorization Concepts, Access Control, Secure API Design, Data Integrity, System Reliability, Production Debugging, Threat-Aware Engineering

AI & Machine Learning

AI Systems, Machine Learning, Computer Vision, TensorFlow, PyTorch, AI Model Integration, Scientific Computing

Infrastructure & Development

Linux, Docker, GitLab CI/CD, Git, CMake, Unit Testing, Integration Testing, Test Automation

Data & Processing

SQL, Relational Databases, Data Processing, Data Pipelines, Data Validation, ETL Concepts, Distributed Data Processing

Education

M.Sc. Scientific Computing, University of Vienna (Oct. 2013–Jun. 2017)

Master Thesis: Experimental Studies of the Influence of Silent Data Corruption on FT-GMRES

Focus: Algorithms, Numerical Methods, High Performance Computing

B.Sc. Industrial Simulation, University of Applied Sciences St. Pölten (Sep. 2010–Jun. 2013)

Open Source & Projekte

Photo Archive: Docker-basierte Anwendung für Bildverwaltung [GitHub](#)

Game of Life: C++ Simulation mit Fokus auf Algorithmen und Softwaredesign [GitHub](#)

Rust Game of Life: Rust Implementierung [GitHub](#)

Domino Simulation: C++ Simulation [GitHub](#)

Sprachen

Deutsch – Muttersprache

Englisch – Fließend in Wort und Schrift

Soft Skills

- Structured analytical thinking and effective problem solving for complex technical challenges
- Independent ownership of technical tasks and pragmatic decision-making
- Commitment to sustainable software development with focus on quality and maintainability
- Effective collaboration in interdisciplinary and international teams